D-Modem

AT Command Reference

The modem core of dmodemd, libdmodem and the PBX modules

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1 Introduction

D-Modem's modem core (`core/` in the source) takes the AT commands of every D-Modem program:

- **dmodemd**, the modem on a pseudo terminal (`/dev/ttyDM0` links to it), its line a SIP call;
- **libdmodem** (`dmodem.h`), the modem as a library: the program that embeds it sends the commands;
- the **Asterisk and FreeSWITCH modules** (`app_dmodem`, `mod_dmodem`), whose profiles give the commands each call starts with.

The command set is D-Modem's own. It began as Smart Link's slmodemd's (the reference of Smart Link's ST56 modems is kept as `docs/legacy/ST56ATCommands.pdf`) and has grown V.90 and V.92 as the digital modem, V.91, V.44, V.92's modem-on-hold, and MNP 2 to 5 since. It takes as well the commands that other modems' setup strings send (V.250's, Rockwell's and Conexant's, USRobotics'), most of them with no effect, so that a setup string written for another modem works: section 10 lists them, and where D-Modem's meaning of a command differs from another modem's. Section 9 tells what a SIP line changes in dmodemd.

This reference follows the core as it is: every command, parameter range, default and output here is the core's own. A setting that is kept but changes nothing says so.

Notation. `AT` commands are shown in capitals; the modem takes either case. *n* is a decimal number, *h* a hexadecimal one; `[ ]` marks what may be left out. A setting's default is the one `ATZ` and the start of the program give.

2 Command lines

- A command line begins with `AT` and ends with the character of S3 (carriage return) or S4 (line feed). Blanks and tabs in it are dropped before it is read.
- The character of S5 (backspace) takes back the last one typed.
- A line holds 63 characters at most; a longer one is ERROR.
- With echo on (`E1`), what the DTE types comes back as it is typed.
- A line that does not begin with `AT` is not a command: with echo on it is blanked out on the DTE's screen.
- `A/` (no line end needed) runs the last command line again.

The commands of a line run in turn, one after the other (`ATE1Q0V1`); a basic command's number left out is 0 (`ATE` is `ATE0`). An extended command (`+...`) takes =*values*, `?` (its values) and `=?` (its ranges). A `;` may stand between two commands, as V.250 puts one after an extended command that another follows (`AT+IFC=2,2;+MS=V92`); here it is not needed. The first command that fails ends the line with ERROR; otherwise OK follows the last one. `A`, `D` and `O` end the line and give their result later (CONNECT, NO CARRIER and so on), and `Z` ends it with OK: whatever follows them on the line is not read, but for a dial string's `;`, after which the rest of the line runs once the number is dialed (`ATD`). A command that reports something (`I`, `S`*n*`?`, `+MS?` ...) prints it on lines of its own before the result.

A character from the DTE while a call is being set up (dialing, answering, the modems' training) hangs up; after a dial string's `;` the modem is in command state, and takes commands.

3 Result codes

V0	V1	
0	OK	the command line ran
1	CONNECT	the call is up, or a fax command's data begins
n	CONNECT *rate*	a data call up at *rate* (below)
2	RING	a call rings (dmodemd: every 3 s, section 9)
3	NO CARRIER	no call came up, or the call is over
4	ERROR	a command failed, or a line too long
6	NO DIALTONE	no dial tone (X2, X4)
7	BUSY	a busy tone
8	NO ANSWER	no answer within S7 seconds of dialing
88	FAX	adaptive answer (+FAA=1): a fax call, class 1 follows
89	DATA	adaptive answer: a data call, its CONNECT follows

Q1 turns them off, V0 gives the numbers, each on a line of its own. A data call's CONNECT gives the rate the modem receives at, in bit/s (CONNECT 33600), and with V0 that rate's number (23); with X0, in fax and in voice mode it is CONNECT (1) alone. \V1 adds the error correction and the compression to it (CONNECT 33600/LAPM/V42bis). The rates' numbers are Smart Link's (its ST56 manual's), and D-Modem's own from 80 for the rates that has not; a rate with neither is 1:

rate	n	rate	n	rate	n	rate	n
75	80	24000	19	37333	51	50666	58
300	9	25333	81	38000	43	52000	32
600	10	26400	20	38666	52	53333	59
1200	5	26666	82	40000	25	54000	33
2400	11	28000	46	41333	53	54666	60
4800	12	28800	21	42000	26	56000	34
7200	13	29333	47	42666	54	57333	83
9600	27	30666	48	44000	28	57600	35
12000	14	31200	22	45333	55	58000	44
14400	15	32000	40	46000	29	58666	84
16800	16	33333	49	46666	56	60000	45
19200	17	33600	23	48000	30	61333	85
21600	18	34666	50	49333	57	62666	86
		36000	42	50000	31	64000	87

Before CONNECT, &A (S70) adds lines on the connection, each bit one of them; with V0 the first three are numbers, Smart Link's, and the fourth is left out:

bit	line	V0
0 (1)	Modulation: *name*, the modulation (V92, V34, V32bis, ...)	1000 and its number (1092, 1034, 1132, ...)
1 (2)	Protocol: LAPM, MNP or NONE, the error correction (V.42's LAPM, MNP 2 to 4)	77, 78, 76
0 and 1 (3)	Compression: V44, V42bis, MNP5 or NONE, what V.42 or MNP agreed	68, 67, 66, 69
2 (4)	TxRate: *rate*, the rate the modem sends at	none

4 Basic commands

ATA

Answers: off hook, and the answering modem's tones after S92 seconds. In dmodemd the call that rings (RING) is answered. Its result is CONNECT, or NO CARRIER. On a call held after a dial string's ; it answers that call (V.250's call-back: ATD*number*;A dials, then answers).

ATB, ATB*n*

ATB alone: V.22 at 1200 bit/s only, without automode, AT+MS=22,0,1200,1200. B0 to B3, B15 and B16, other modems' choice between the ITU and the Bell modulation at 1200 and 300 bit/s: taken, changing nothing (the call's negotiation and +MS choose).

ATD*string*

Dials *string* and calls. The dialer sends the digits and * # A B C D as DTMF (S11 ms each, as long between them); T and P choose tone or pulse dialing (S16) for the rest of the call. Pulses are dialed on an analogue line (dmodemd's -a and -usb), with the country's break, make, gap between digits and digit pattern (+GCI; * # A B C D skipped); a line that cannot pulse, such as SIP, dials them in DTMF. These modifiers do as V.250 has them:

,	a pause of S8 seconds
W	waits for a dial tone (1 s of it): NO DIALTONE if none comes in S7 seconds
@	waits for the remote end's ringing (0.5 s of sound at least), then for 5 s of quiet after it: NO ANSWER if that does not come in S7 seconds
;	ends the dial string: once the number is dialed the modem is back in command state, off hook with the call kept, and says OK; the rest of the line runs then

Other characters (!, the hook flash, among them: a SIP call has no hook to flash) are skipped; S=*n* (a stored number) is ERROR, as there are none. ATDL dials the last number again (Dialing *number*... first). With X2 or X4 the modem waits for a dial tone first. The result is CONNECT, BUSY, NO ANSWER, NO DIALTONE or NO CARRIER.

After a ; the call waits, silent, for the DTE: ATD*more* dials more on it (as DTMF; a ; again holds it again), ATD alone or ATO starts the calling modem's handshake, ATA the answering modem's, and ATH hangs up (OK, then NO CARRIER). ATDL is ERROR then. In dmodemd the dial string is the SIP call's number (section 9).

ATE[*n*]

Echo of the DTE's characters in command mode: E0 off, E1 on (S13; default 1).

ATH[0], ATH1

H or H0 hangs up (on hook); H1 goes off hook.

ATI[*n*]

Information:

I0	the product: D-Modem
I1, I2	the product and version (D-Modem, 1.0.0), then the maker, D-Modem project
I3	the modem's name, then its line's kind (dmodemd: dmodem0, SIP)
I4	S0 to S23, eight a line (s00=000 s01=000 ...)
I5, I6	Stored Profile 0:, 1: (there are none)
I7	the country: Country: USA
I11	the last call's statistics, or this one's (below)

Any other number is ERROR.

`ATI11` lists the last call's statistics, as Conexant's modems do: in online command state those of the call that is up, after a call (dialed or answered, connected or not) its own, until the next one; before any, `No Previous Call`. A line whose value the call does not have is left out (the rates and what follows for a call that never came up, the frame counts without error correction, the SNR, the round trip delay and the echoes where the modulation does not measure them, the cause while the call is up):

line	
`Last Connection`	the modulation it came up with (`V34`, `V90`, `V92`, `V91`, `V32bis`, ...), `Fax`, or `None`
`Initial ..., Final Transmit / Receive Carrier Rate`	the rates (bit/s) at CONNECT, and now or at the end (after renegotiations)
`Protocol Negotiation Result`	`LAPM`, `MNP` or `NONE`
`Data Compression Result`	`V44`, `V42bis`, `MNP5` or `NONE`
`Estimated SNR (dB)`	the receiver's signal to noise ratio (V.34, V.90's upstream, V.32bis)
`Receive / Transmit Signal Power Level (dBm0)`	the line's mean power each way while the call was up
`Round Trip Delay (ms)`	as V.34's phase 2 measured it
`Near / Far Echo Level (dBm0)`	the echoes of our signal the canceller takes away (V.34, V.90, V.92)
`Characters Sent / Received`	the DTE's
`Transmit Frame Count`	LAPM's or MNP's frames sent
`Transmit Frame Error Count`	the information frames among them sent again
`Receive Frame Count`	the frames received intact
`Receive Frame Error Count`	the frames received with a bad FCS
`Retrains by Local Modem, Retrains Requested by Remote Modem`	retrains, ours and the other modem's
`Rate Renegotiation by Local / Remote Modem`	rate renegotiations (V.34, V.90, V.92)
`Call Duration`	from CONNECT, *h*:*mm*:*ss*
`Call Termination Cause`	`Local hangup` (`ATH`, `ATZ`, DTR, a character while connecting), `Remote cleardown` (the other modem's LAPM DISC or MNP LD), `Line dropped` (the SIP call ended), `Carrier lost`, `No dial tone`, `Busy`, `No answer`, `Training failed`, `Protocol failure` (no error correction where it is required, LAPM's errors), `No carrier`

```
Description                     Status
-----------                     ------
Last Connection                 V34
Initial Transmit Carrier Rate   33600
...
Call Duration                   0:00:10
Call Termination Cause          Local hangup
```

A digital modem has no view of the line's digital loss or robbed-bit signalling (the analogue modem measures them), so those other modems' lines are not there.

`ATL[n]`, `ATM[n]`

The speaker's volume, `L0` to `L3` (`S30`; default 3), and when it is on, `M0` never, `M1` until the call is up, `M2` always, `M3` as `M1` (on other modems not while dialing) (`S29`; default 1). A SIP line has no speaker: kept for the commands that set them. A SmartUSB56 with `dmodemd -usb-buzzer` sounds its buzzer while the speaker is on (at any volume but `L0`).

ATN[*n*]

Automode (S31; default 1): N1 lets the modems find their best common modulation from the one +MS chose down, N0 keeps to it. The same as +MS's second parameter.

ATO[*n*]

Back online after the escape (section 8): CONNECT. On a call held after a dial string's ;, the calling modem's handshake (as ATD). ERROR when there is no call to go back to. O0 only goes back; O1 then retrains (V.250's) and O3 renegotiates the rate (Rockwell's), both in V.34, V.90 and V.92 calls (in the others it only goes back); O2 and the other numbers are ERROR. The retrain or renegotiation counts as the local modem's in ATI11.

ATP, ATT

Pulse or tone dialing (S16; default tone): see ATD.

ATQ[*n*]

Q0 gives the result codes, Q1 keeps them back (S14; default 0).

ATS*n*?, ATS*n*=*v*

Reads S-register *n* (0 to 255), in decimal on a line of its own, or sets it to *v* (8 bits: *v* modulo 256). A number over 255 is ERROR. Section 7 lists them.

ATV[*n*]

V0 gives the result codes as numbers, V1 as words (S15; default 1).

ATX[*n*]

Results and dial tone, X0 to X4 (S56; default 4):

	CONNECT	before dialing
X0	without the rate	no wait for a dial tone
X1, X3	with the rate	no wait for a dial tone
X2, X4	with the rate	a dial tone of 1 s (steady, or in a cadence whose gaps are 1 s at most, as Italy's: its tone added up),

BUSY on a busy tone comes in X3 and X4 only, as in the Hayes set: the others take no notice of a busy tone. A busy tone is four cycles of the country's busy or congestion tone's cadence (the ITU's list's, 15% or 80 ms either side: 0.48 s on and off or 0.24 in Germany, 0.75 s in India, 1 s on and 4 off in Egypt), not its ringing tone; a country with neither (none of the 192) its table's bounds. NO ANSWER comes in all of them, and the dial string's W and @ wait in all of them. A SIP call has no dial tone: dmodemd wants ATX3.

ATY[*n*]

Y0 to Y3: kept in S161, nothing more.

ATZ[*n*]

Z or Z0 to Z3: hangs up, and the S-registers, +MS, modem-on-hold, the data mode (+FCLASS=0) and the parameters kept for other modems' setup strings (section 10) as at the start; the country stays. There are no stored profiles: the number changes nothing. OK, and the rest of the line is not read.

5 Ampersand, backslash and percent commands

AT&A[*n*]

&A0 to &A7: the lines before CONNECT (S70; default 0), see the result codes.

AT&C[*n*], AT&E[*n*], AT&H[*n*]

Data carrier detect (&C0, &C1; S60), the rate CONNECT gives (&E0, &E1; S71) and flow control (&H0 to &H3; S62): kept, but they change nothing (a pty has no carrier line, CONNECT always gives the receive rate, and flow control is the pty's). Default 0.

AT&D[*n*], AT&F[*n*], AT&R[*n*], AT&S[*n*]

Taken, with or without a digit, and nothing done (ATZ gives the defaults).

AT%C[*n*], AT&K[*n*]

Data compression (S104; default 3): %C0 none, %C1 MNP 5 (with MNP), %C2 V.44 or V.42bis both ways (with V.42's LAPM, as the other modem agrees: V.42 offers both), %C3 either, as the call's error correction has it. It needs error correction (\N). &K is compression as on USRobotics' modems: &K0 none, &K1 and &K2 both (%C3), &K3 V.44 or V.42bis without MNP 5 (%C2); Rockwell's and Conexant's use it for flow control, whose &K4 to &K6 are taken with no effect, and whose &K3 (RTS/CTS) leaves V.42's compression on.

AT&P[*n*]

&P0 to &P3: the pulse dialing ratio, kept in S28.

AT&Q[*n*]

Other modems' choice of error correction, as \N: &Q0 and &Q6 are \N0 (none), &Q5, &Q8 and &Q9 are \N3 (V.42, then MNP, then none). The others (synchronous modes) are ERROR.

AT&V[0], AT&W[*n*]

&V and &V0 show the active settings, in lines of 72 columns at most: the basic commands', the S-registers in use (S*nn*:*vvv*), +MS and the other extended settings, and the parameters kept for other modems' setup strings (section 10). There are no stored profiles: the list ends with STORED PROFILES: none, and &W, &W0, &W1 (store one) answer OK, storing nothing. &V1 and up are ERROR.

```
ACTIVE PROFILE:
E1 L3 M1 N1 Q0 T V1 X4 Y0 &A0 &C0 &E0 &H0 &P0 \A3 \N3 \V0 %C3 %E0
S00:000 S01:000 S02:043 S03:013 S04:010 S05:008 S06:002 S07:060 S08:002
...
+MS=92,1,300,56000 +FCLASS=0 +GCI=b5 +PMH=0 +PMHT=5 #CID=0 +IFC=2,2
...
STORED PROFILES: none
```

AT\A[*n*]

MNP's block size, the longest information field it offers (S105; default 3): \A0 64, \A1 128, \A2 192, \A3 256 octets (as the other modems do, an LR offering 64 and the 256-octet data phase optimization). \A4 and up are ERROR.

AT\N[*n*]

Error correction (S103; default 3), as Smart Link's and Rockwell's modems have it; each sets +ES to V.250's equal:

		+ES
\N0	normal: none	1,0,1
\N1	direct: none	0,0,0
\N2	reliable: V.42 (LAPM) or MNP, else hang up	3,2,4
\N3	auto-reliable: V.42, MNP, or none	3,0,2
\N4	V.42 (LAPM) only, else hang up	3,3,5
\N5	MNP only, else hang up	4,4,6

V.42's detection phase decides between LAPM and none; the originator sends MNP's LR when the answerer sends no V.42 answer, and the answerer takes an LR (from a modem with MNP only, or one set so) as the start of MNP. With MNP only the originator starts with its LR. MNP 5 compression goes with MNP (%C), MNP's block size is \A's. ATS103=*n* is \N*n*.

AT\V[*n*]

\V1 (\V2 the same): a data call's CONNECT has the error correction and the compression after its rate, CONNECT 33600/LAPM/V42bis, CONNECT 2400/MNP/MNP5, CONNECT 28800/NONE; \V0 the rate alone (S106; default 0). With V0 or X0 it changes nothing.

AT%E[*n*]

%E0 to %E2, line quality monitoring: kept in S39 (the modulations retrain and renegotiate their rate by themselves).

AT*B, AT%B

The numbers barred from dialing: none are kept; OK.

AT#CID=*n*, AT#CID?, AT#CID=?, AT+VCID...

Caller ID between the first two rings: 0 off, 1 or 2 on (=? gives 0,1,2). 1 gives the fields (DATE = , TIME = , NMBR = , NAME = , a line each), 2 the whole message in hex (MESG =). +VCID (V.253's) is the same, in every mode. In voice mode (+FCLASS=8) the lines come as V.253's complex event report, between <DLE>X and <DLE>.; in the other modes after an empty line. A SIP call rings without it.

6 Extended commands

6.1 Modulation: +MS

`AT+MS=[mod][,[automode][,[min][,[max]]]]`

The modulation the modem calls or answers with, whether it may fall back (*automode* 1, as `N1`; `S31`) and the rates it may connect at, *min* to *max* bit/s (300 to 56000, *min* no more than *max*). *mod* is a number or V.250's name for it, in either case (`AT+MS=V34` is `AT+MS=34`); `?` and `=?` give the numbers. A parameter left out keeps its value: `AT+MS=,0` only turns automode off. The default is `92,1,300,56000`. In V.34 *max* caps both directions' rates (sent in its MP sequences).

mod	**name**		
21	`V21`	ITU V.21	300 bit/s FSK
23	`V23C`	ITU V.23	1200/75 bit/s FSK
103	`B103`, `Bell103`	Bell 103	300 bit/s FSK
22	`V22`	ITU V.22	1200 bit/s DPSK
212	`B212`, `Bell212A`	Bell 212A	1200 bit/s DPSK
122	`V22B`	ITU V.22bis	2400 bit/s, V.22 as its fallback
32	`V32`	ITU V.32	4800 and 9600 bit/s
132	`V32B`	ITU V.32bis	4800 to 14400 bit/s
34	`V34`	ITU V.34	2400 to 33600 bit/s, after V.8
90	`V90`	ITU V.90	as the digital modem, after V.8
91	`V91`	ITU V.91	two digital modems, after V.8
92	`V92`	ITU V.92	V.90 with quick connect, PCM upstream and modem-on-hold
56	`K56`	K56flex	not implemented: runs V.90
150	`V_CVM`	V.cvm	data over mobile speech paths, 175 to 3500 bit/s, after V.8 (`+CVM`)

V.90 and V.92 are there as the *digital* modem (the server's side, as on a G.711 VoIP line), on an 8000 Hz G.711 line: analogue V.90 and V.92 modems calling connect at up to 56000 bit/s downstream. 90 and 92 offer V.91 too, which two digital modems use instead, and V.34 for the others; on a line at another sample rate they are V.34. With automode V.8 and the V.34 modem hand over to V.32bis (and its V.22bis and V.22) for a modem without V.34, and the answering modem tries V.23 and Bell 103 for one without V.8.

`V_CVM` is the proposed Recommendation of `docs/v_cvm.pdf`, for calls whose speech path is a mobile codec's (AMR, AMR-WB), which carry neither V.34 nor V.32: V.8 with V.22 and V.21 as the fallback, then a probing of the codec and a mode loaded for it. The start-up takes 15 to 21 s; `S7` should allow for it. The connection runs LAPM.

`AT+MS?`

The settings: `92,1,300,56000`.

`AT+MS=?`

What may be chosen:
`(21,22,23,122,32,132,34,103,212,56,90,91,92,150),(0,1),(300-56000),(300-56000)`

6.2 Identification and country

`AT+GMI, AT+GMM, AT+GMR`

The manufacturer (`D-Modem project`), the model (`D-Modem`) and the revision (`1.0.0`, the build's version); with `?` after them (`AT+GMI?`) the same.

`AT+GCAP`

The command sets the modem has: `+GCAP: +FCLASS,+MS`.

`AT+GCI=`*h*`, AT+GCI?, AT+GCI=?`

The country by its T.35 code in hexadecimal (`+GCI:b5`: USA), which sets its dial pause (S8) and hook flash (S24) times; `=?` lists the codes. dmodemd's `-c` option names the country it starts with (`-countrylist` lists them): one of the table's, or any of the 192 of the ITU's list of the tones of national networks (after E.180), by its name there in capitals or the table's (`-c FIJI`, `-c UNITED_KINGDOM`: `UK`). The table's countries have their own dialling settings; the others the USA's (`ATI7`: `Country: FIJI (USA line)`), and all of them their own busy and congestion tones.

6.3 Error correction: +ES

`AT+ES=[`*orig*`][,[`*orig_fb*`][,[`*ans_fb*`]]], AT+ES?, AT+ES=?`

V.250's error control selection (default `3,0,2`): as the originator, *orig* 0 or 1 none, 2 V.42 (LAPM) without the detection phase, 3 V.42 with it, 4 MNP first (no detection phase, no LAPM), 6 V.80's Synchronous Access Mode (section 6.4), and *orig_fb* what may follow when that fails: 0 or 1 none, 2 V.42 or MNP or hang up, 3 V.42 only, 4 MNP only (and first); as the answerer, *ans_fb* 0 or 1 none, 2 or 3 V.42, MNP or none, 4 V.42 or MNP or hang up, 5 V.42 only, 6 MNP only, 8 Synchronous Access Mode. V.80's Synchronous Mode and Frame Tunnelling Mode (5 and 7, 7 and 9) are not there. A parameter left out keeps its value; `=?` gives the ranges `(0-4,6),(0-4),(0-6,8)`. `\N` sets it, and it sets S103 to the `\N` with its values (as it was if none has them).

6.4 V.80 Synchronous Access Mode: +ESA, +ITF

With `+ES=6` (calling) or `+ES=,,8` (answering) the call has no error correction or compression: after CONNECT the modem moves the DTE's octets as a synchronous bit stream, with V.80's in-band commands after EM (19h) both ways (Table 9/V.80). It starts in the transparent sub-mode: the DTE's octets go out as they are, least significant bit first, and the line's bits come back as octets; when the DTE has nothing the modem sends *syn1* and tells it `<EM><under>`. With `+ESA`'s *trans_idle* 1 or 2 the receiver first hunts for the SYN, and gives the DTE nothing before it. `<EM><flag>` from the DTE starts the framed sub-mode: HDLC frames, each ended by the DTE's `<EM><flag>`, the modem adding the zero insertion, the flags and the FCS; it gives back each received frame's octets (without the FCS it checked) and `<EM><flag>` for a good one or `<EM><err>` for a bad one, an `<EM><err>` when its hunt finds a flag, and `<EM><mark>` for an abort or marks. `<EM><mark>` from the DTE goes back to transparent. EM and 99h in the data are shielded both ways (`<EM><t1>`, `<EM><t2>`; DC1 and DC3 too with DC1/DC3 flow control, `+IFC`); from the DTE all the codes T1 to T20 are taken.

The other commands: `<EM><ecs>` (instead of +++, which is data) gives `<EM><ecs>`, then OK in online command state, where `ATO` goes back; `<EM><eot>` gives `<EM><eot>` and hangs up (NO CARRIER); `<EM><rate><tx><rx>` sets the highest rates to send at (`+MS`'s *max*) from *tx* and to receive at from *rx*, for the next retrain or renegotiation (`AT+MS` sets both again); `<EM><hunt>` restarts the receiver's hunt (framed, or transparent with *trans_idle* 1 or 2), what waits for the DTE dropped; `<EM><resume>` after an underrun (below). `<EM><rtn>` retrains the call and `<EM><rrn>` renegotiates its rate, as `ATO1` and `ATO3`. The modem gives `<EM><eot>` before NO CARRIER when the remote's carrier goes, `<EM><rtn>` and `<EM><rrn>` for each retrain and rate renegotiation but the DTE's own (the remote's, or the modem's after a loss of signal), `<EM><rate><tx><rx>` after each, or when the rates change in the call, `<EM><rover>` where received octets were lost (the DTE took nothing for 2048 octets), `<EM><tover>` where its own were (the buffer full), and `<EM><bnum>` with `+ITF`. V.34's half duplex is not there.

`AT+ESA=[`*trans_idle*`][,`*framed_idle*`...][,`*syn2*`], AT+ESA?, AT+ESA=?`

The Synchronous Access Mode's eight settings, each left out as it is (default `0,0,0,0,0,0,255,255`): *trans_idle*, the transparent sub-mode's SYN, 0 *syn1* sent when the DTE has nothing and the receiver's octets as they come, 1 the same but the receiver hunting for *syn1*, 2 *syn1* and *syn2* sent in turn and the receiver hunting for the two (*syn1* first), each hunt giving the DTE the octets from the SYN on, aligned on it; *framed_idle* between frames 0 flags, 1 marks; *framed_un_ov* when the DTE runs out in a frame, 0 an abort, 1 a flag (no FCS), the DTE's octets then dropped until its `<EM><resume>`, which `<EM><unum>` and their count answer (both: `<EM><under>`); *hd_auto* 0; *crc_type* 0 no FCS, 1 FCS-16, 2 FCS-32 (V.42's); *nrzi_en* 0; *syn1* and *syn2* as above. `=?` gives `(0-2),(0,1),(0,1),(0),(0-2),(0),(0-255),(0-255)`.

`AT+ITF=[`*off*`][,[`*on*`][,[`*report_period*`]]], AT+ITF?, AT+ITF=?`

The Synchronous Access Mode's transmit buffer (default `383,128,0`): the DTE's octets in it, EM codes included, stop being taken above *off* (*on* is kept; *off* must be above it, else ERROR), and with *report_period* not 0 the modem gives <EM><bnum> with their number every *report_period* times 10 ms. `=?` gives `(1-4095),(0-4094),(0-500)`.

6.5 Mode

`AT+FCLASS=`*n*`, AT+FCLASS?, AT+FCLASS=?`

`0` data (default), `1` fax class 1, `8` voice (`=?` gives `0,1,8`).

`AT+FAA=`*n*`, AT+FAA?, AT+FAA=?`

Adaptive answer. (In voice mode, see section 6.9: a calling tone on the call switches it to fax or data.) With `1`, a call the modem answers (`ATA`, S0) in data or fax mode is answered as a data call (its answer tone, V.8's ANSam, serves a fax machine as its CED), and found out to be a fax machine's or a data modem's:

- a fax call by the caller's CNG (1100 Hz, 0.5 s of every 3.5 s; or a burst cut short as the answer tone starts, a fax machine that stops its CNG on hearing it), the V.8 CM of a fax call (a call function of facsimile, as V.34 fax machines and some class 1 modems send it on ANSam), the far end going over to T.38 (dmodemd's SIP calls), or a caller silent through every answer signal of the automode chain (ANSam, then V.32's, V.22's, V.21's, V.23's and Bell 103's, some 20 s, within T.30's T1 of 35 s): FAX, `+FCLASS` becomes 1, and the modem goes on as a class 1 answerer: what is left of CED's 3 s, then CONNECT, as after `AT+FTH=3`, for the DTE's first frames;
- a data call by its modem's handshake: DATA when it comes up, then its CONNECT; `+FCLASS` becomes 0.

A call that ends with neither leaves `+FCLASS` as it was. A V.22, V.21 or Bell 103 caller, silent until it hears its own answer signal, answers its stage of the chain: DATA. `0` (default) answers as `+FCLASS` says. `=?` gives `+FAA: (0,1)`; `Z` sets 0. Rockwell's modems give FAX and DATA too, as 33 and 35 with `V0`: here those are rates' CONNECTs, so 88 and 89.

6.6 V.92 modem-on-hold

`AT+PMH=`*n*`, AT+PMH?, AT+PMH=?`

`0` lets the other modem put the call on hold, `1` refuses (default 0; `+PMH?` gives `+PMH: 0`).

`AT+PMHT=`*n*`, AT+PMHT?, AT+PMHT=?`

How long a hold the modem grants: the code of V.92's Table 33, `0` to `13` (`0` grants none; default 5).

`AT+PMHR[=`*s*`]`

In online command mode on a V.92 connection: asks the other modem to go on hold, for *s* seconds (1 to 960; 10 if left out). ERROR otherwise.

6.7 V.cvm: +CVM

`AT+CVM=[`*modes*`][,[`*rate*`][,[`*report*`]]], AT+CVM?, AT+CVM=?`

For `+MS=V_CVM`: *modes*, the modes allowed, a sum of 1 (the robust mode), 2 (narrowband), 4 (wideband), 8 (8-DPSK), 16 (the amplitude rings) and 32 (the parametric mode); default 63, all of them. *rate* is the signal interface's rate, 8000 or 16000 (default 8000); *report*, 0 or 1 (default 0), asks for `+CVMR` on each renegotiation. D-Modem keeps *rate* and *report* but works at 8000 Hz only, without the wideband modes, and sends no `+CVMR`. `+CVM?` gives `+CVM: 63,8000,0` and, during and after a V.cvm call, a second line with that call's figures: `+CVM:` *raw tx*`,`*raw rx*`,`*coded tx*`,`*coded rx*`,`*symbol ms*`,`*tones tx*`,`*tones rx*`,`*code rate*`,`*renegotiations*`,`*erasures* (`+CVM: 1600,1600,1400,1400,20.0,16,16,7/8,0,0`; the tones are 0 in the parametric mode). `=?` gives `+CVM: (0-63),(8000,16000),(0,1)`. The call's CONNECT has its coded receive rate and `/CVM`: `CONNECT 1200/CVM`, with `\V1 CONNECT 1200/CVM/LAPM`. S44 and S45 hold the tones loaded and the renegotiations.

6.8 Fax class 1

With AT+FCLASS=1, the modem takes T.31's class 1 commands, the fax machine's side being the DTE's (the PBX modules and libdmodem have a T.30 engine of their own as well). +FAA=1 answers fax and data calls alike (above); +FPR is kept with no effect (section 10). As the caller, the modem sends CNG after dialing until the answerer's flags come in its first +FRH=3 (T.30 has it until the DIS).

AT+FMI?, AT+FMM?, AT+FMR?

The manufacturer, the model and the revision, as +GMI, +GMM and +GMR give them (in any mode).

AT+FTS=*n*, AT+FRS=*n*

Sends silence for, or waits for silence of, *n* times 10 ms.

AT+FTM=*m*, AT+FRM=*m*, AT+FTM=?, AT+FRM=?

Sends or receives data at modulation *m*: CONNECT, then the data with <DLE> doubled, ended by <DLE><ETX>.

m	
24, 48	V.27ter, 2400 and 4800 bit/s
72, 96	V.29, 7200 and 9600 bit/s
73, 97, 121, 145	V.17, 7200 to 14400 bit/s, long training
74, 98, 122, 146	V.17, 7200 to 14400 bit/s, short training

AT+FTH=3, AT+FRH=3, AT+FTH=?, AT+FRH=?

Sends or receives HDLC frames at V.21 channel 2 (300 bit/s), the only one: =? gives 3.

6.9 Voice

With AT+FCLASS=8 the modem takes these voice commands (ERROR in the other modes):

command	
+VIP	the voice settings as at the start
+VNH=*n*	0 or 1
+VGT=*n*, +VGR=*n*	the gain sending and receiving, 0 to 128
+VIT=*n*	the inactivity timer, 0 to 255
+VSD=*s*,*t*	silence detection: sensitivity and period, 0 to 255
+VSM=*m*,8000	the compression method, 0 or 1, at 8000 samples/s
+VLS=*n*	1 takes the line for voice (on SIP: answers the call that rings), 0 leaves it; 14 and 15 a handset's (below). =? lists them, each with the events it reports (below)
+VEM=*n*	0 or 1: V.253's event reports in the speakerphone state, which the modem has not; kept, 0 at the start and with +VIP
D*number*	dials *number* and gives OK when the far end answers, BUSY, NO ANSWER after S7, or ERROR when the call fails otherwise. An answer tone is the answer and a busy tone BUSY on any line (on SIP in its early media too); on SIP the answer is otherwise the 200 OK, on a line that does not tell the ringing tone's end (+VRA), or +VRN without one. With the line taken it dials the digits as DTMF, S11 ms each
H	leaves the line and goes back to data mode, +FCLASS=0

command	
+VRA=*n*	the far end has answered when the ringing tone has been gone this long since its last ring, in 100 ms, 0 to 255 (0: as the first ring ends); 50 at the start
+VRN=*n*	the far end has answered when no ring has come in this long after dialling, in seconds, 0 to 255 (not beyond S7); 10 at the start
+VTX, +VRX, +VTR	sends, receives, or both at once: CONNECT, then 8-bit samples at 8000/s, <DLE> doubled. +VTX ends at <DLE><ETX>, <DLE>! or <DLE><CAN> from the DTE, +VRX at any character from it, +VTR at <DLE>^ only; then OK, after <DLE><ETX> to end the samples of +VRX and +VTR. A gap in the DTE's samples plays as silence
+VTD=*d*	the duration of a +VTS element that gives none, 0 to 6000 (0: 1 s); 100 at the start
+VTS=*string*	plays the elements of *string*, separated by commas, in turn, and gives OK after the last; ERROR, and nothing played, if one cannot be. An element is:
s	the DTMF digit *s* (0–9, *, #, A–D), or ! a hook flash
{*s*,*d*}	the DTMF digit or hook flash *s* for *d*
[*f1*,*f2*,*d*]	a tone of *f1* and *f2* Hz (0 to 4000; 0 is none) for *d*
	Durations are in 10 ms, 0 to 6000. A field may be left empty: a frequency is then 0, a digit none (silence), and a duration +VTD's, so {5}, [440] and [] last +VTD and {} is silence for as long. Two digits in a row get 50 ms of silence between them. +VTS=1,2,3 dials 123. A hook flash is the line's: over SIP an RFC 4733 event (16), the audio silent meanwhile; a string with ! is an ERROR on a line that has none (a SIP call whose far end takes no telephone-events).

Each + command takes ? (its values) and =? (their ranges). +VCID, caller ID, is #CID's, in every mode.

While the line is taken (+VLS=1) the modem listens for DTMF digits from the far end and reports them as V.253 has it: <DLE> and the digit (0–9, *, #, A–D) when one is heard, 51 ms into it, and again every 70 ms while it lasts, then <DLE>~ when it ends. The reports come in the voice command state as well as among the samples of +VRX, whose audio is silence from a digit's report until it ends. The modem's own +VTS tones are not reported.

The modem also reports, in every state, each ring of a ringing tone as <DLE>r (a tone of 400 to 480 Hz, or 440 and 480 Hz, or 400 and 450 Hz, on for 0.6 s at least; the UK's two bursts are one ring) and a busy or congestion tone as <DLE>b (four rounds of 0.2 to 0.8 s on and off), and an answer tone as <DLE>a (2100 Hz: a fax or data answer) or <DLE>f (2225 Hz: a data answer), 0.3 s into it. A ring ends after 0.3 s without the tone. A calling fax machine's CNG (1100 Hz, bursts of 0.36 to 0.8 s) is <DLE>c, a calling data modem's V.25 tone (1300 Hz) <DLE>e, each as a burst ends and 4 s apart at least, as V.253 asks.

AT+VLS=? gives V.253's list of the devices, with the events each reports in the transmit, receive and command states as hex bit fields (bit *n* for V.253's event *n*, event 0 the leftmost of 32):

```
0,"",00000000,00000000,90000000
1,"T",0B043060,0B043060,0B043060
```

On hook (0) that is caller ID (with +VCID) and RING; on the line (1) DTMF, the two calling tones, the loop current interruption, a ringing tone, a busy tone and the two answer tones, in every state. A line that senses the phone on the modem's own jack (a SmartUSB56 with `dmodemd -usb-handset`) adds event 8 to both, in every state (00800000, 0B843060, ...): that phone off hook is <DLE>H, on hook again <DLE>h. A modem with a handset of its own (a SmartUSB56 with `dmodemd -usb-voice-codec`) lists two more:

```
14,"H",08000000,08000000,08000000
15,"HT",0B043060,0B043060,0B043060
```

+VLS=14 puts the voice engine on the handset alone, the line on hook: +VRX records its microphone, +VTX plays to its earpiece, and DTMF from it is reported. +VLS=15 takes the line with the handset on the call: the DTE hears the line

and the handset, the line gets the DTE's samples and the handset. Moving between 1, 14 and 15 keeps the voice engine running, the line taken or left.

A voice call can become a fax or data call (V.253 5.3.2 and its example I.6). +FCLASS=1 (or 0) in the voice command state, off hook, keeps the call: the voice engine stops, the line stays off hook and silent, and A answers on it as a fax (class 1: CED, then CONNECT for the DIS) or data modem, or D calls on it (call progress waits for the far end's answer tone). H, Z or the far end hanging up ends the held call. For example, a fax machine calls while the greeting plays:

```
AT+VTX                  CONNECT      the greeting's samples follow
                        <DLE>c       the caller's CNG
<DLE><ETX>              OK
AT+FCLASS=1             OK
ATA                     CONNECT      class 1, as after AT+FTH=3
```

With +FAA=1 the modem does it itself, in the voice command state (a +VTX, +VRX, +VTR or +VTS under way ends first): after <DLE>c, FAX, +FCLASS=1 and the fax answer's CONNECT; after <DLE>e, +FCLASS=0 and the adaptive data answer (DATA, then its CONNECT). +VNH is accepted and has no effect: a held call's failed handshake hangs up.

When the far end ends a voice call, the modem reports <DLE>l (V.253's loop current interruption, "usually a remote hangup") and goes on hook (+VLS=0); V.253 has no NO CARRIER in voice mode. What was under way ends: +VRX's samples with <DLE><ETX>, then OK; +VTX and +VTR when the DTE ends them (its samples until then are discarded), then OK, +VTR's after <DLE><ETX>; +VTS's tones with ERROR. In the command state <DLE>l comes alone.

7 S-registers

Registers 0 to 255 hold 8 bits each. Those not listed are 0 and free; “kept” means set and read back, with no other effect.

S		default	
0	rings to answer after (0: never)	0	
1	rings so far	0	
2	the escape character	43 (+)	
3	the line’s end	13 (CR)	
4	the line feed	10 (LF)	
5	backspace	8 (BS)	
6	wait for a dial tone, s	2	X2, X4; 2 s at least
7	wait for the answer and the training, s	60	
8	a comma’s pause, s	2	the country’s
9	carrier detect time, 0.1 s	6	kept
10	carrier loss before hanging up, 0.1 s	7	
11	DTMF tone and gap, ms	100	below 40: 95
12	escape guard time, 20 ms	50	kept: the guard is 0.5 s
13	echo (E)	1	
14	quiet (Q)	0	
15	verbose (V)	1	
16	tone (1) or pulse (0) dialing (T, P)	1	
24	hook flash time, 10 ms	20	the country’s
28	pulse ratio (&P)	0	kept
29	speaker control (M)	1	
30	speaker volume (L)	3	
31	automode (N, +MS)	1	
32	the modulation (+MS)	92	
39	line quality (%E)	0	kept
44	V.cvm: the tones loaded in the receive direction of the last call (+CVM)	0	read only
45	V.cvm: the last call’s renegotiations	0	read only
56	results and dial tone (X)	4	
60	carrier detect (&C)	0	kept
62	flow control (&H)	0	kept
70	CONNECT’s lines (&A)	0	
71	CONNECT’s rate (&E)	0	kept
92	the answer’s delay, s (0: 10 ms)	2	before the answer tone
103	error correction (\N)	3	
104	data compression (%C, &K)	3	
105	MNP’s block size (\A)	3	
106	the protocol in CONNECT (\V)	0	
161	Y	0	kept

8 Online, the escape and hanging up

Online, what the DTE sends goes to the other modem, except the escape: three S2 characters (+++), each within 0.5 s of the one before, with 0.5 s of nothing before them and after them. The modem then says OK and is in online command mode, the call still up: `ATO` goes back online, `ATH` hangs up, and on a V.92 connection `AT+PMHR` asks for modem-on-hold. The +++ itself is data (the modem cannot know it is the escape until the guard time after it), and what the DTE sent before the escape and the line had not taken yet is kept, not dropped: it goes to the other modem, +++ last, once `ATO` is back online.

The call ends with NO CARRIER when the other side hangs up or its carrier is gone for S10 tenths of a second. dmodemd hangs up as well when the DTE drops DTR (it sets the pty's speed to 0) or, with HUPCL set on the pty, closes it; it then makes a new pty (`/dev/ttyDM0` follows it).

9 dmodemd on a SIP line

- `ATD`*number* calls `sip:`*number*`@`*domain* of `SIP_LOGIN`. A telephone number (only V.250's dial characters: digits, `* # + A B C D`, the modifiers `T P W , @ !` `;` and `- ( ) .` and blanks) goes as its digits and DTMF characters: `ATDT555-1000` calls `sip:5551000@`*domain*. Anything else is a SIP user's name, as it is: `ATDalice`.
- The line has no dial tone: `ATX3` (or `X1`), else NO DIALTONE.
- A dial string with a `;` (`ATDT5551000;`) makes the SIP call and says OK at once, the modem in command state: what `ATD` adds then goes as DTMF on the call (a menu's choices, an extension), and `ATD` alone or `ATA` starts the modem's handshake.
- With `-L` dmodemd registers and takes calls: RING every 3 seconds while one waits, `ATA` or `S0` rings (`ATS0=1`) answer it, and the rings stop if the caller gives up. A call that comes while the line is in use is refused (486, busy); without `-L` every call is (480).
- The call is G.711 at 8000 Hz, which is what V.90 and V.92 need (`AT+MS=92`, the default). Over a real PSTN gateway a lower modulation (`AT+MS=132,0,4800,9600`) may do better.

10 Setup strings from other modems

A setup string written for another modem (a program's, a driver's, a guide's) holds commands of V.250, of Rockwell's and Conexant's modems, of USRobotics' and of Hayes'. The modem takes the common ones: those it has a meaning for do what the sections above say, and the others, below, are taken with no effect, so that the line ends in OK. Flow control is the pty's own and the line a G.711 call, and the modulations retrain and renegotiate by themselves: what these commands set elsewhere has no use here.

10.1 Basic commands taken

Each with the values listed (a number left out is 0), OK and nothing more; another value is ERROR.

command	on other modems
`B0 to B3, B15, B16`	the ITU or the Bell modulation at 1200 and 300 bit/s (`ATB` alone: V.22 here)
`C1`	the carrier on (`C0`, receive only, is ERROR)
`F1`	no echo of the data online (`F0` is ERROR)
`W0 to W2`	what CONNECT reports (here `&A` and `&E`)
`&B0 to &B2`	V.32's retrain; USRobotics': the DTE's rate fixed
`&G0 to &G2`	V.22bis's guard tone
`&I0 to &I5`	USRobotics' XON/XOFF flow control
`&J0, &J1`	the auxiliary relay
`&K4 to &K6`	Rockwell's flow control (`&K0 to &K3`: compression here)
`&M0, &M4, &M5`	asynchronous mode; USRobotics': error correction (here `\N`)
`&N0 to &N14`	USRobotics': the line's rate (here `+MS`)
`&W0, &W1`	store the profile (there are none)
`&Y0, &Y1`	the profile at power on
`\G0, \G1`	XON/XOFF passed to the other modem
`\J0, \J1`	the DTE's rate following the line's
`\K0 to \K5`	what a break does
`\Q0 to \Q3`	flow control
`\R0, \R1`	the ring indicator after the answer
`\T0 to \T255`	the inactivity timer
`-C0, -C1`	the calling tone
`-K0 to -K2, -Q0, -Q1, )M0 to )M2, @M0 to @M30, *H0 to *H2`	Rockwell's MNP 10 and 10EC (cellular): MNP Extended Services, the V.22bis fallback, power levels, the link's speed

`&D, &F, &R, &S` are taken as well (section 5).

10.2 Extended parameters kept

These keep their values with no other effect: =*values* sets them (each in its range, else ERROR; one left out stays as it is), `?` gives +*name*: *values* (`-SEC?` the values alone), `=?` the ranges, and `ATZ` the defaults.

name	default	on other modems
`+IFC`	`2,2`	flow control, each way (0 to 3, 0 to 2; here DC1/DC3 in V.80's Synchronous Access Mode, section 6.4)
`+ICF`	`3,3`	the DTE's character framing
`+IPR`	`0`	the DTE's rate (0 automatic, 300 to 921600)

name	default	on other modems
+ILRR	0	the DTE's rate reported
+IDSR	0	DSR
+ILSD	0	long space disconnect
+MR	0	the modulation reported (+MCR:, +MRR:)
+ER	0	the error correction reported (+ER:)
+DR	0	the compression reported (+DR:)
+DS	3,0,2048,32	V.42bis (here %C)
+DS44	3,0,0,2048,2048, 255,255,6144,6144	V.44 (here %C)
+PCW	2	V.92's call waiting (none is detected here)
+PIG	1	V.92's PCM upstream, an analogue modem's (this one is the digital modem)
+FPR	0	the fax DTE's rate
-SEC	0	Rockwell's MNP 10EC, and its transmit level (1,12)

10.3 Where D-Modem differs

- &K is the compression, as on USRobotics' modems; on Rockwell's and Conexant's it is flow control. &K3 leaves V.42's compression on (MNP 5 off), &K0 turns it off.
- The S-registers past S16 are D-Modem's own: S29 and S30 are the speaker's (elsewhere the hook flash time and the inactivity timer), S92 the answer's delay (elsewhere a fax transmit level: ATS92=10 makes the modem wait 10 s before its answer tone). The others (S36, S37, S48, S91 . . .) are free: kept as written. S7 and S11 start at 60 and 100.
- The dial string's ! (hook flash) and $ (bong tone) are skipped. There are no stored numbers: &Z and ATDS=*n* are ERROR.
- ATI's numbers are the modem's own (I0 to I7, and I11 as Conexant's); ATI8 to ATI10 and ATI12 and up are ERROR.
- The numeric results (V0) are Smart Link's, CONNECT's for each rate among them; those of USRobotics' and Conexant's modems differ. CONNECT's rates without a number of Smart Link's have D-Modem's own, 80 to 87.
- MNP is classes 2 to 5 (\N, %C): there is no MNP 10 or 10EC. A modem with MNP 10EC connects with V.42 (its "EM" detection patterns are passed over). See docs/link.md, MNP.
- K56flex is not there: +MS=56 runs V.90.

10.4 Not there

ERROR: &V1 and up, &Z (no stored numbers), &T (self tests), \B (a break), X5 and up, -V90 (+MS chooses), +MA, +MSC, +EB, +ESR, +ETBM, +EWIND, +EFRAM, +A8E, +A8T, +PMHF, +GSN, +GOI and #UD; +MS's names of modulations the modem has not got (V32T, V29 . . .).